A Spine Isolation Theorem for Magma Implications

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Abstract

We classify the 4,694 magma equations of the Equational Theories Project by spine type, the path from the parse-tree root to the leftmost occurrence of the distinguished variable, into pure left-spine, pure right-spine, mixed, trivial, and product-LHS families. We prove that a pure left-spine equation of depth n cannot imply any equation with an incompatible spine: neither right-spine, nor mixed-spine, nor left-spine of depth m where $n \nmid m$. Parts (b) and (c) extend the first-letter invariant of Corollary 10.4 in the Equational Theories Project blueprint to a systematic spine-type classification. Part (a) introduces a new family of separating magmas, cyclic groups with the successor operation, yielding a complete divisibility characterization. The theorem is validated with zero exceptions on the full 22-million-pair Lean-verified implication graph, resolving 1,540,946 ordered pairs. The core separation arguments are formally verified in Lean 4, with no Mathlib dependency.

1 Introduction

The Equational Theories Project (ETP) of Tao et al. [1] catalogues 4,694 magma equations and determines, via Lean-verified proofs, the complete implication graph among them: 22,033,636 ordered pairs, each labelled TRUE (implication holds in every magma) or FALSE. The resulting structure (1,415 equivalence classes connected by 4,824 edges in the transitive reduction) is rich enough to motivate a search for systematic separation principles.

One such principle appears in the ETP blueprint [2], Chapter 10 ("Some abstract nonsense"). Corollary 10.4 defines the *first letter of* w invariant: for any magma word w, the leftmost leaf of the term tree is preserved by the left-zero magma $M = (S, \cdot)$ where $a \cdot b = a$ for all $a, b \in S$. If two equations E_1 and E_2 differ on this invariant (that is, if the leftmost leaf of the right-hand side of E_1 is the left-hand-side variable x, while the leftmost leaf of E_2 is not) then the left-zero magma satisfies E_1 but not E_2, proving $E_1 \not\Rightarrow E_2$.

This paper extends the first-letter invariant in two directions. First, we introduce a spine-type classification that partitions the equations into pure left-spine, pure right-spine, and mixed-spine families (plus trivial and product-LHS families outside the theorem's scope). Parts (b) and (c) of our main theorem show that the left-zero magma separates not only by the identity of the leftmost leaf but by the entire spine type: pure left-spine equations cannot imply pure right-spine or mixed-spine equations. Second, Part (a) introduces a new family of separating magmas (the cyclic groups $\mathbb{Z}/n\mathbb{Z}$ equipped with the successor operation $a \cdot b = a + 1 \pmod{n}$) that stratifies equations within each spine type by depth, yielding the complete divisibility characterization: depth n implies depth m only if $n \mid m$.

The theorem applies when E_1 has a pure left-spine (or, by a mirror construction, a pure right-spine)

and E_2 has a bare variable as its left-hand side. It does not apply when E_2 belongs to Family C, where the left-hand side is itself a product.

2 Definitions and Classification

Definition 1 (Spine)**.** *Let E be a magma equation of the form $x = t$, where t is a magma term over the variable set containing x. The* spine *of E is the sequence of directions $d_1 d_2 \cdots d_k \in \{L, R\}^*$ obtained by recording, at each binary operator node from the root of t to the leftmost occurrence of x, whether x lies in the left (L) or right (R) subtree.*

Definition 2 (Pure left-spine of depth n)**.** *An equation $x = t$ has a* pure left-spine of depth n *if the spine is L^n: the leftmost occurrence of x in t is reached by taking the left child at every node from the root, traversing exactly n operator nodes. Equivalently, the right-hand side has the form $(\cdots((x \cdot T_1) \cdot T_2) \cdots T_n)$ where $T_1, \ldots, T_n$ are arbitrary magma terms.*

Definition 3 (Pure right-spine; mixed spine)**.** *An equation has a* pure right-spine of depth n *if the spine is R^n. An equation has a* mixed spine *if the spine contains both L and R steps.*

We classify the 4,694 equations of the ETP catalog [1] (reproduced in the accompanying repository [3]) by spine type. The distribution is as follows.

Spine type	Count	Depth 1	Depth 2	Depth 3	Depth 4
Pure left	815	297	295	171	52
Pure right	240	88	87	50	15
Mixed	1,267	—	—	—	—
Trivial	817	—	—	—	—
Family C	1,555	—	—	—	—

The pure left-spine and pure right-spine families admit a natural depth grading. The remaining 3,639 equations (mixed, trivial, and Family C) fall outside the scope of the present theorem.

3 Main Result

Theorem 1 (Spine Isolation)**.** *Let E_1 be a magma equation of the form $x = (\cdots((x \cdot T_1) \cdot T_2) \cdots T_n)$, so that the leftmost leaf of the right-hand side is x, reached by taking the left child at every node from the root, a pure left-spine of depth $n \geq 1$. The variable x may appear elsewhere in the right-hand side. Let E_2 be any magma equation whose left-hand side is a bare variable. All separating magma arguments below require $|S| \geq 2$. Then:*

(a) If E_2 has a pure left-spine of depth m where $n \nmid m$, then E_1 does not imply E_2.

(b) If E_2 has a pure right-spine, then E_1 does not imply E_2.

(c) If E_2 has a mixed spine, then E_1 does not imply E_2.

Corollary 1 (Right-Spine Isolation)**.** *Let E_1 have a pure right-spine of depth n, and let E_2 have a pure right-spine of depth m with $n \nmid m$. Then E_1 does not imply E_2.*

Remark 1. *The divisibility condition $n \mid m$ is necessary but not sufficient for implication between same-spine equations. On the ETP catalog, the fraction of TRUE implications among divisible*

left-spine depth pairs ranges from 24% to 39%, confirming a substantial gap between the necessary condition and sufficiency.

4 Proofs

4.1 Proof of Parts (b) and (c)

Proof. We construct a separating magma that satisfies every pure left-spine equation but fails every pure right-spine and every mixed-spine equation.

Let S be any set with $|S| \geq 2$, and define the *left-zero magma* $M = (S, \cdot)$ by $a \cdot b = a$ for all $a, b \in S$.

M **satisfies every pure left-spine equation.** Let E_1 be a pure left-spine equation of depth n: $x = (\cdots((x \cdot T_1) \cdot T_2) \cdots T_n)$. We show by induction on depth that the right-hand side evaluates to x under every assignment in M.

Base case ($n = 1$): The right-hand side is $x \cdot T_1$. In M, $x \cdot T_1 = x$.

Inductive step: Suppose the claim holds for depth $n-1$. The right-hand side of a depth-n equation is $(\cdots((x \cdot T_1) \cdots T_{n-1}) \cdot T_n)$. By the inductive hypothesis, the inner expression $(\cdots((x \cdot T_1) \cdots T_{n-1}))$ evaluates to x. Then $x \cdot T_n = x$ in M, completing the induction.

M **fails every pure right-spine equation.** Let E_2 be a pure right-spine equation of depth k: $x = (T_1 \cdot (T_2 \cdot \ldots (T_k \cdot x) \ldots))$. In M, the outermost operation evaluates to $T_1 \cdot (\text{anything}) = v(T_1)$, the value of T_1 under the given assignment v. Since E_2 has a pure right-spine, the leftmost leaf of the right-hand side is the leftmost leaf of T_1, which is not x (the leftmost leaf is x only for left-spine equations). In M, every term evaluates to its leftmost leaf, so $v(T_1)$ equals the leftmost leaf of T_1. Choose an assignment sending this leaf to some $a \in S$ with $a \neq x$ (possible since $|S| \geq 2$). Then the right-hand side evaluates to $a \neq x$, so E_2 fails in M.

M **fails every mixed-spine equation.** Let E_2 have a mixed spine. Following the spine from the root, at some node we take the right child for the first time. At that node the left-zero operation discards the right subtree (which contains the leftmost occurrence of x) and retains the left subtree, which does not contain x along the spine path. The right-hand side therefore evaluates to a term whose value is independent of x (it equals the leftmost leaf of the left subtree at that node). Choosing an assignment where this value differs from x gives a counterexample.

Since M satisfies E_1 but fails E_2 in all cases, E_1 does not imply E_2. □

4.2 Proof of Part (a)

Proof. We construct, for each depth n, a magma that satisfies every pure left-spine equation of depth n but fails every pure left-spine equation of depth m whenever $n \nmid m$.

Define the *cyclic magma* $M_n = (\mathbb{Z}/n\mathbb{Z}, \cdot)$ by $a \cdot b = a + 1 \pmod{n}$.

k-fold left-composition equals $a + k \pmod{n}$. We verify by induction. For $k = 1$: $a \cdot T_1 = a + 1$. Assuming the $(k-1)$-fold left-composition equals $a + (k-1)$, the k-fold composition is $(a + (k-1)) \cdot T_k = (a + (k-1)) + 1 = a + k \pmod{n}$. Note that the right operand T_k does not affect the result: M_n discards the right operand in the same way as the left-zero magma, except that it increments the left operand by one.

Depth-n equations hold in M_n. A pure left-spine equation of depth n evaluates to $x + n \equiv x \pmod{n}$. Hence $M_n \models E_1$.

Depth-m equations fail in M_n when $n \nmid m$. A pure left-spine equation of depth m evaluates to $x + m \pmod{n}$. When $n \nmid m$, we have $x + m \not\equiv x \pmod{n}$, so the equation fails for every $x \in \mathbb{Z}/n\mathbb{Z}$.

Since M_n satisfies E_1 (depth n) but not E_2 (depth m, $n \nmid m$), the implication $E_1 \Rightarrow E_2$ fails. □

4.3 Proof of the Right-Spine Corollary

Proof. Given a magma $(S, \cdot)$, define the *opposite magma* $M^{\mathrm{op}} = (S, \cdot')$ by $a \cdot' b = b \cdot a$. A pure right-spine equation of depth n under $\cdot$ becomes a pure left-spine equation of depth n under $\cdot'$: the sequence of R-steps in the original parse tree becomes a sequence of L-steps when left and right operands are exchanged at every node.

Apply Theorem 1(a) to M^{op}: if E_1 has a pure left-spine of depth n under $\cdot'$ (equivalently, a pure right-spine of depth n under $\cdot$) and E_2 has a pure left-spine of depth m under $\cdot'$ (a pure right-spine of depth m under $\cdot$) with $n \nmid m$, then E_1 does not imply E_2 in M^{op}. Since M^{op} is a magma, this witnesses $E_1 \not\Rightarrow E_2$. □

5 Empirical Validation

We validated the theorem on the complete ETP implication graph: 4,694 equations, 22,033,636 ordered pairs, each resolved by Lean-verified proof. The spine classification was computed by a C parser operating on the equation parse trees. All parts of the theorem produce zero violations.

5.1 Part (a): Non-divisible left-spine depths

Table 1 shows all eight non-divisible depth pairs among pure left-spine equations. Every pair yields zero TRUE implications, confirming Part (a).

Table 1. Non-divisible left-spine depth pairs ($n \nmid m$). All implications are FALSE.

Source depth	Target depth	$n \nmid m$	Pairs	TRUE
2	1	$2 \nmid 1$	87,615	0
2	3	$2 \nmid 3$	50,445	0
3	1	$3 \nmid 1$	50,787	0
3	2	$3 \nmid 2$	50,445	0
3	4	$3 \nmid 4$	8,892	0
4	1	$4 \nmid 1$	15,444	0
4	2	$4 \nmid 2$	15,340	0
4	3	$4 \nmid 3$	8,892	0

Table 2 shows the divisible depth pairs, where implication is permitted by the theorem. The nonzero TRUE rates (24–39%) confirm that the divisibility condition is tight: replacing $n \nmid m$ with a weaker condition would introduce false negatives.

Table 2. Divisible left-spine depth pairs ($n \mid m$). Implication is possible; the theorem does not apply.

Source depth	Target depth	$n \mid m$	TRUE	FALSE	Rate
1	2	$1 \mid 2$	21,966	65,649	25.1%
1	3	$1 \mid 3$	12,355	38,432	24.3%
1	4	$1 \mid 4$	3,726	11,718	24.1%
2	4	$2 \mid 4$	6,021	9,319	39.3%

5.2 Parts (b) and (c): Cross-spine isolation

Every pure left-spine → pure right-spine pair and every pure left-spine → mixed-spine pair yields zero TRUE implications across approximately 1,228,000 ordered pairs.

Table 3. Cross-spine pairs from pure left-spine sources.

Source type	Target type	Pairs (approx.)	TRUE
Pure left	Pure right	195,600	0
Pure left	Mixed	1,032,605	0

5.3 Right-spine corollary

Table 4 confirms the right-spine analogue. All eight non-divisible right-spine depth pairs yield zero TRUE implications.

Table 4. Non-divisible right-spine depth pairs ($n \nmid m$). All implications are FALSE.

Source depth	Target depth	$n \nmid m$	Pairs	TRUE
2	1	$2 \nmid 1$	7,656	0
2	3	$2 \nmid 3$	4,350	0
3	1	$3 \nmid 1$	4,400	0
3	2	$3 \nmid 2$	4,350	0
3	4	$3 \nmid 4$	750	0
4	1	$4 \nmid 1$	1,320	0
4	2	$4 \nmid 2$	1,305	0
4	3	$4 \nmid 3$	750	0

We note an asymmetry in the catalog: pure right-spine → pure left-spine implications are 14–17% TRUE, not universally FALSE. This is because the 4,694-equation catalog is not closed under mirroring: 554 of the 4,694 equations (predominantly Family C, with product left-hand sides) have no mirror image in the catalog. The cross-spine isolation (Parts (b) and (c)) is a statement about left-spine sources; it does not assert the converse. Within any mirror-closed subset of equations, full left-right symmetry would hold.

In total, the theorem resolves **1,540,946 ordered pairs** with zero violations.

6 Formal Verification

The core separation arguments of all three parts of Theorem 1 are formally verified in Lean 4 (v4.29.0) with no Mathlib dependency [3]. The formalization comprises three modules totalling

approximately 300 lines.

Definitions (`SpineIsolation/Defs.lean`). An inductive type `MagmaTerm` represents magma terms over natural-number variables, with constructors `var` and `op`. The predicates `IsLeftSpine x n t` and `IsRightSpine x n t` are defined inductively, capturing the spine classification of Section 2. A structure `MagmaEq` pairs a left-hand-side variable with a right-hand-side term; `MagmaEq.holds` quantifies universally over assignments and carriers, while `MagmaEq.notImplies` witnesses separation existentially.

Parts (b) and (c) (`SpineIsolation/LeftZero.lean`). The key lemma `leftZero_eval_eq_leftmostVar` establishes that in the left-zero magma, evaluating any term yields the assignment at its leftmost leaf variable. The proof proceeds by structural induction on `MagmaTerm`. The main theorem `leftSpine_notImplies_nonLeftSpine` constructs the separating witness on `Bool`: assigning `true` to the left-hand-side variable and `false` to all others.

Part (a) (`SpineIsolation/Cyclic.lean`). The lemma `cyclic_eval` proves that in the cyclic magma on `Fin n`, evaluating a term t under assignment f yields $f(\text{leftmostVar}(t)) + \text{leftSpineDepth}(t) \pmod{n}$. The main theorem `leftSpine_depth_notImplies` combines this with the witness $f = \lambda_.\ 0$ to separate depth n from depth m when $n \nmid m$.

The formalization is self-contained: no external libraries beyond Lean's core are required. All proofs are constructive, producing explicit separating magmas rather than asserting their existence.

7 Discussion

The divisibility characterization of Part (a) is sharp in one direction: $n \mid m$ is necessary for $E_1 \Rightarrow E_2$ when both equations are pure left-spine. The gap between necessary and sufficient is substantial: even when $n \mid m$, only 24–39% of pairs in the catalog are TRUE. Closing this gap, perhaps by enriching the separating magma family, is an open problem.

The left-right asymmetry observed in the empirical data is a catalog artifact, not an algebraic phenomenon. The ETP catalog enumerates equations by a left-to-right tree-structure convention, producing 554 equations (predominantly Family C) with no mirror image in the catalog. Within any mirror-closed extension, the right-spine analogue of Parts (b) and (c) would hold with identical force.

More broadly, the results illustrate a methodology: structural features of equation parse trees (here, the spine path) can serve as algebraic separation witnesses when paired with appropriately chosen magma families. The left-zero magma and the cyclic successor magmas $(\mathbb{Z}/n\mathbb{Z}, a \cdot b = a + 1)$ are elementary objects. The contribution is not the magmas themselves but the systematic classification of which equations they separate.

References

[1] T. Tao et al., "The Equational Theories Project: Advancing Collaborative Mathematical Research at Scale," *arXiv:2512.07087*, 2025.

[2] T. Tao et al., "Equational Theories Project Blueprint," https://teorth.github.io/equational_theories/blueprint/, 2025. Chapter 10, "Some abstract nonsense," Corollary 10.4.

[3] A. McKenna, “Spine Isolation Theorem: Lean 4 Formalization,” https://github.com/mysticflounder/equational-magma-theorems, 2026.